

Software Process & Quality Management

Wordle Game – Console Application

1. Team Information

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Project Overview

• Project Description

The Wordle Game System is an interactive word guessing application designed to provide users with an engaging and entertaining experience. In this system, the user attempts to guess a hidden word within a limited number of attempts. The system randomly selects a word at the beginning of each game session.

After each guess, the system evaluates the user's input and provides feedback indicating whether each letter is correct and in the correct position, correct but in the wrong position, or not included in the word. This feedback helps guide the user toward the correct answer. The game continues until the user successfully guesses the word or exhausts all allowed attempts. The system then displays the final result and gives the user the option to start a new game.

• Objectives

The main objectives of this project are:

1. To design and implement a functional word guessing game based on the Wordle concept.
2. To provide users with an engaging and enjoyable gameplay experience.
3. To ensure the system is simple, intuitive, and easy to use.
4. To implement an accurate feedback mechanism that assists users in improving their guesses.
5. To apply software engineering principles and follow a structured development lifecycle.

• Key Features

The system provides the following key features:

1. Starting a new game session at any time.
2. Random generation of a hidden word for each game.
3. Allowing the user to input guesses through the interface.
4. Validating user input to ensure it matches the required word length.
5. Providing feedback after each guess, indicating correct and incorrect letters.
6. Limiting the number of attempts to a fixed number (e.g., six attempts).
7. Displaying the final result (win or lose) at the end of the game.
8. Allowing the user to replay the game after completion.

• Target Users

The system is intended for general users who enjoy word-based games and puzzles. It is designed for users with basic computer skills and does not require any prior technical knowledge. The system is suitable for players of different age groups due to its simplicity and ease of use.

Lifecycle Justification

• Selected Model

The Waterfall model has been selected as the software development lifecycle for this project.

• Justification for Choosing Waterfall

The Waterfall model is suitable for this project because the system requirements are clearly defined and stable from the beginning. The Wordle game is a simple application with well-understood functionality and does not require frequent updates or changes during development.

The Waterfall model allows each phase of the project to be completed before moving on to the next phase, ensuring clarity and structure throughout the development process.

Additionally, the Waterfall model emphasizes documentation, which aligns with the academic requirements of this project. Each stage, including requirements, design, implementation, and testing, can be clearly documented and reviewed.

• Development Approach

The project will follow the main phases of the Waterfall model as follows:

- **Requirements Analysis:** In this phase, all system requirements will be identified and documented in the Software Requirements Specification (SRS).
- **System Design:** The system architecture and design diagrams, such as ER diagrams and use case diagrams, will be created.
- **Implementation:** The system will be developed based on the approved design specifications.
- **Testing:** The system will be tested using predefined test cases to ensure that all functional and non-functional requirements are satisfied.
- **Deployment:** The final version of the system will be prepared for demonstration and submission.

Each phase will be completed and reviewed before proceeding to the next phase to ensure a structured and well-organized development process.